

Kubernetes HPA Autoscaling Not Working SOP

Objective

Restore Horizontal Pod Autoscaler behavior when pods do not scale according to CPU, memory, or custom metric thresholds.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Kubernetes cluster using kubectl and the correct namespace context.
2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T025
Issue	Kubernetes autoscaling not working
Category	Scaling
Service	Kubernetes
Severity	High
Status	Resolved
Description	HPA not scaling pods
Expected Resolution	Check metrics server and thresholds

Procedure

1. Confirm HPA name, target deployment, min/max replicas, and configured metrics.
2. Check HPA status and events for missing metrics or failed scale decisions.
3. Validate metrics-server, custom metrics adapter, or Prometheus adapter health.
4. Confirm pods have CPU and memory requests set because HPA needs resource requests for utilization metrics.
5. Review current load and compare it with configured scale thresholds.
6. Check deployment replica constraints, quotas, and pod scheduling capacity.
7. Fix metrics availability, thresholds, resource requests, or scaling limits.
8. Generate controlled load and verify replicas scale up and down as expected.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
HPA not scaling	Metrics unavailable	Fix metrics-server or adapter
CPU utilization unknown	Missing resource requests	Add CPU and memory requests
Scale up blocked	Max replicas or quota reached	Adjust limits or capacity
Scaling delayed	Threshold too high or stabilization window	Tune HPA settings

Best Practices

1. Set realistic resource requests.
2. Monitor HPA current metrics and desired replicas.
3. Load test autoscaling behavior.
4. Use stabilization windows to reduce flapping.

References

[Kubernetes Docs](#) | [Internal Tickets CSV](#) | [Cluster Events](#) | [Monitoring Dashboards](#) | [Application Logs](#) | [Platform Runbook](#)